

Building a highly accurate digital twin of the Earth



Destination Earth

Under the leadership of the European Commission, and in coordination with the Member States, three entrusted entities are developing the main components of the DestinE initiative.



European Space Agency (ESA)

Responsible for DestinE Platform, the user entry to the DestinE System. Based on a multi-cloud infrastructure, the platform will provide access to data, applications and services.



European Centre for Medium-Range Weather Forecasts (ECMWF)

Responsible for the creation of the first two digital twins; Climate Adaptation and Weather-induced Extremes, as well as the the Digital Twin Engine.



European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT)

Responsible for the DestinE Data Lake: built from geographically distributed physical elements, offering discovery and data access to a large data portfolio (digital twins data and more) and big data processing near data.

Margrethe Vestager

Executive Vice-President for a Europe fit for the Digital Age



“Destination Earth will improve our understanding of climate change and enable solutions at global, regional and local levels. This initiative is a clear example that we cannot fight climate change without digital technologies. For example, the digital modelling of the Earth will help to predict major environmental degradation with unprecedented reliability.”

Thierry Breton

Commissioner for the Internal Market



“With Destination Earth, we are building on Europe's strong cards. From Artificial Intelligence, cloud computing, high-speed connectivity networks to our successful Copernicus Earth observation programme and our world-leading EuroHPC supercomputers, we are combining our assets in order to make our future more safe and secure.”

Table of Contents

1. About DestinE	p.5
1.1 DestinE roadmap: Looking ahead	p.7
1.2 Who will benefit from DestinE	p.8
2. DestinE System	p.10
2.1 DestinE Platform	p.12
2.2 Data Lake	p.13
2.3 Digital twins and Digital Twin Engine	p.14
3. Use Cases	p.15
4. The DestinE Community	p.20
4.1 Become a User	p.21
4.2 Become a Technology Provider	p.22
5. Procurement Opportunities	p.23

About DestinE

The digital twin of the Earth supporting the development of climate change mitigation strategies.

The increasing frequency and severity of extreme weather events related to climate change makes the need for improved simulations, at the level where the impacts are felt, a priority, in order to adapt to their social and economic consequences in terms of lives, the environment and property.

Destination Earth (DestinE) is a flagship initiative of the European Commission to develop a highly-accurate digital twin of the Earth system (a digital twin of the Earth) to monitor and simulate natural phenomena and the impact of human activities on our planet.

DestinE unlocks the potential of new possibilities in digital modelling of the complex Earth system by providing breakthroughs in terms of accuracy, local detail, fast access to information and interactivity.

This novel type of information system will use unprecedented amounts of real-world observation data, including data from ESA, ECMWF, EUMETSAT, the Copernicus Data and Information Access Services. In combination with innovative Earth system models, advanced Artificial Intelligence (AI), machine learning and Europe's cutting-edge High Performance Computing (HPC), users will be empowered to access the data stream of the digital twins and adapt it to their own needs, bringing the users to the data to help them set more efficient and actionable climate change adaptation and mitigation measures.

The initiative also represents a key component of the European strategy for data by consolidating access to valuable sources of data across Europe.

DestinE main benefits at a glance:

Perform highly accurate, interactive and dynamic simulations of the Earth system (atmosphere, hydrology, land) to enhance climate change and extreme weather prediction capabilities.

Enable the development and testing of updated scenarios to foster data-driven policy-making supporting Europe's efforts to become the world's first climate-neutral continent by 2050.

Enable the development and testing of climate scenarios to assess the socio-economic impact of existing and potential climate change mitigation strategies on society and the environment.

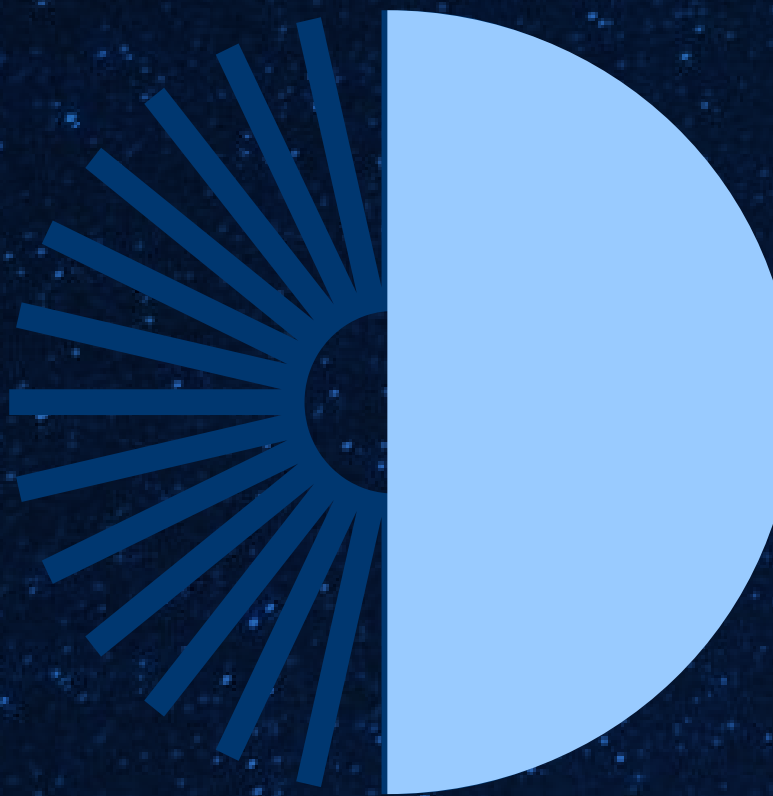
Anticipate extreme weather-driven natural disasters, human-caused environmental damage at the local level, helping anticipate the impacts and reduce life loss and economic downturns.



2022



2023



2024



2026



2030

DestinE Roadmap: Looking ahead

A digital replica of our planet to revisit our past, understand our present and predict our future.

The European Commission is joining forces with the scientific and industrial excellence of the European Space Agency (ESA), the European Centre for Medium-Range Weather Forecasts (ECMWF) and the European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT) to gradually develop the Destination Earth system through the following key milestones:

● **Official launch (2022)**

Official launch of the DestinE initiative.

● **Development (by 2023)**

Development of the three main components begins:

- Destination Earth Service Platform (DESP) implemented by ESA.
- Destination Earth Data Lake implemented by EUMETSAT.
- Digital twins and Digital Twin Engine implemented by ECMWF.

● **Implementation (by 2024)**

The entire system will be made operational.

This phase will focus on configuring, deploying and demonstrating the initial infrastructure that will support DestinE in future phases and the first two digital twins developed by ECMWF:

- Weather-Induced Extremes: This digital twin provides forecasts at unprecedented levels of quality at global and local scales, for close to real-time response to extreme events, as well as an on-demand component allowing user interaction and trial of different scenarios.
- Climate Change Adaptation: This digital twin will provide enhanced simulations at the global and local scales and scenario-simulating capabilities to predict the impact of climate change with increased quality supporting decision makers and the impact sectors for the development of adaptation strategies face to climate change.

● **Completion (by 2026)**

Addition of more digital twins covering different domains of the Earth system (oceans, biodiversity, etc.) and completion of the first two digital twins, the DestinE platform and the data lake and related features to further improve the DestinE system.

● **Integration of digital twins (by 2030)**

All the available thematic digital twins will be gradually integrated to form a comprehensive digital twin of the Earth system.

Who will benefit from DestinE

Digital modelling to move from forecasting what the weather will be, to what the weather will do.

DestinE will enable stakeholders to better understand the environmental and human impacts on Earth, as a complement to the work of the national meteorological and emergency services and European services such as Copernicus. Through a co-design strategy implemented throughout the process, they'll help tailor the system to their real needs.

Industry

Informing industrial operations and identifying opportunities for innovation and growth

All innovative industries will benefit. Energy firms may use weather tech for supply-demand balance; agriculture companies to adapt to new climate conditions; insurance and finance to assess climate-related risks; transportation and construction firms to focus on infrastructure resilience; and consumer goods companies to adapt to sustainability trends and supply chain challenges.



Technology providers

Supplying technology providers with the resources needed to innovate their products and services

Tech providers, including GIS, AI, and cloud computing firms, may offer tools like data processing and analytics to support DestinE's development. This engagement opens new markets and allows for product testing and development.



Decision Makers

Giving decision makers the insights they need to make green and digital transition strategic decisions

Decision makers in government and business may find DestinE appealing for its strategic guidance on sustainability, environmental issues, and digital transformation. The DESP's economic benefits like increased productivity and job creation could also attract them.

Researchers

Advance scientific knowledge and our understanding of the Earth

Researchers and academics can use DestinE data to advance scientific understanding of Earth. The Digital twins' data provides tools to explore complex interactions between the environment and humanity, opening new research avenues.

Policymakers

Simulations and forecasts to enable data-driven and more effective policy decisions

Policymakers at various levels could use Destination Earth data for informed climate policy. DestinE offers analytical tools aligning with the EU's carbon neutrality goals. Local to global policymakers, including city councils, regional authorities, national ministries, and UN representatives, can benefit from data-driven strategies for climate mitigation and adaptation.

DestinE System

The data generated by the DestinE digital twins is included in the DestinE Data Lake and is accessible by users through the DestinE Platform

Overall the DestinE initiative aims to support the development of digital twins of the Earth system to provide decision-makers with accurate and timely information to manage the planet's resources more effectively. As part of this effort, the DestinE System will be made up of three key components including the DestinE Platform, the Data Lake and the Digital Twin Engine.

One of the central components of DestinE initiative is the DestinE Platform. The Platform provides a centralised location for accessing and managing the various services and resources that make up the DestinE ecosystem. It includes a range of tools and capabilities designed to support scientific research, data analysis, and decision-making across a wide range of disciplines. The Platform's key feature is its ability to provide users with on-demand access to a wide range of data sources, including satellite imagery, climate models, weather predictions, and other environmental data.

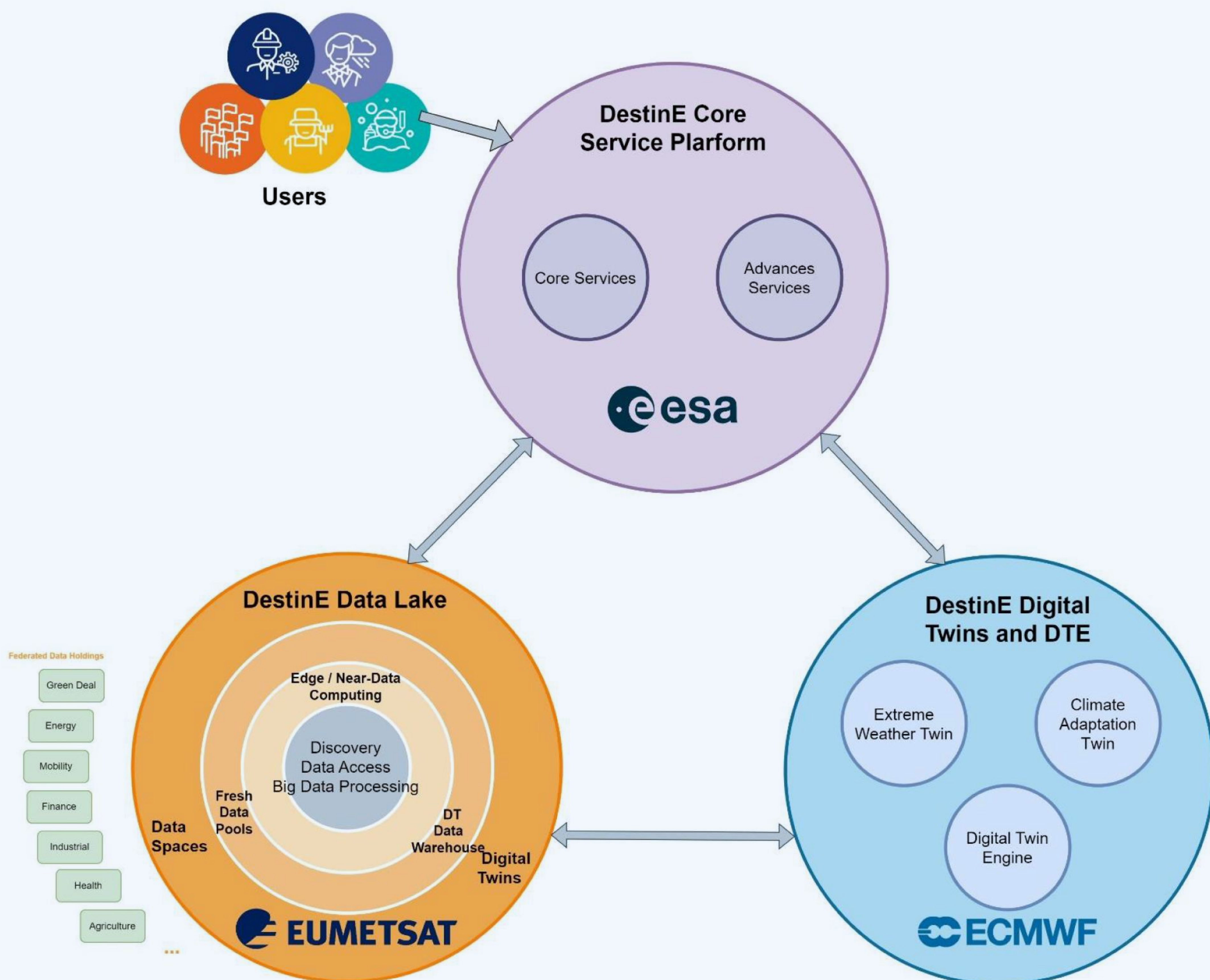


DestinE System

The Data Lake is another crucial component of the DestinE system. The Data Lake will be a large, centralised repository of environmental data that can be accessed by researchers and decision-makers to support scientific research, data analysis, and decision-making across a wide range of disciplines.

The Data Lake's key feature will be its ability to provide users with on-demand access to a wide range of environmental data sources, including for example satellite imagery, climate models, and other environmental data.

Finally, the digital twins and the Digital Twin Engine are central to the entire system. A Digital twin is a virtual representation of an Earth system that can be used to simulate and predict how the environment will respond to various changes or interventions. The Digital Twin Engine is a powerful computational engine that is capable of handling large and complex simulations of the Earth system. Together, the digital twins and Digital Twin Engine allow users (via the DESP) to test various scenarios and evaluate the potential impacts of different policies and interventions.



DestinE Platform

What is it?

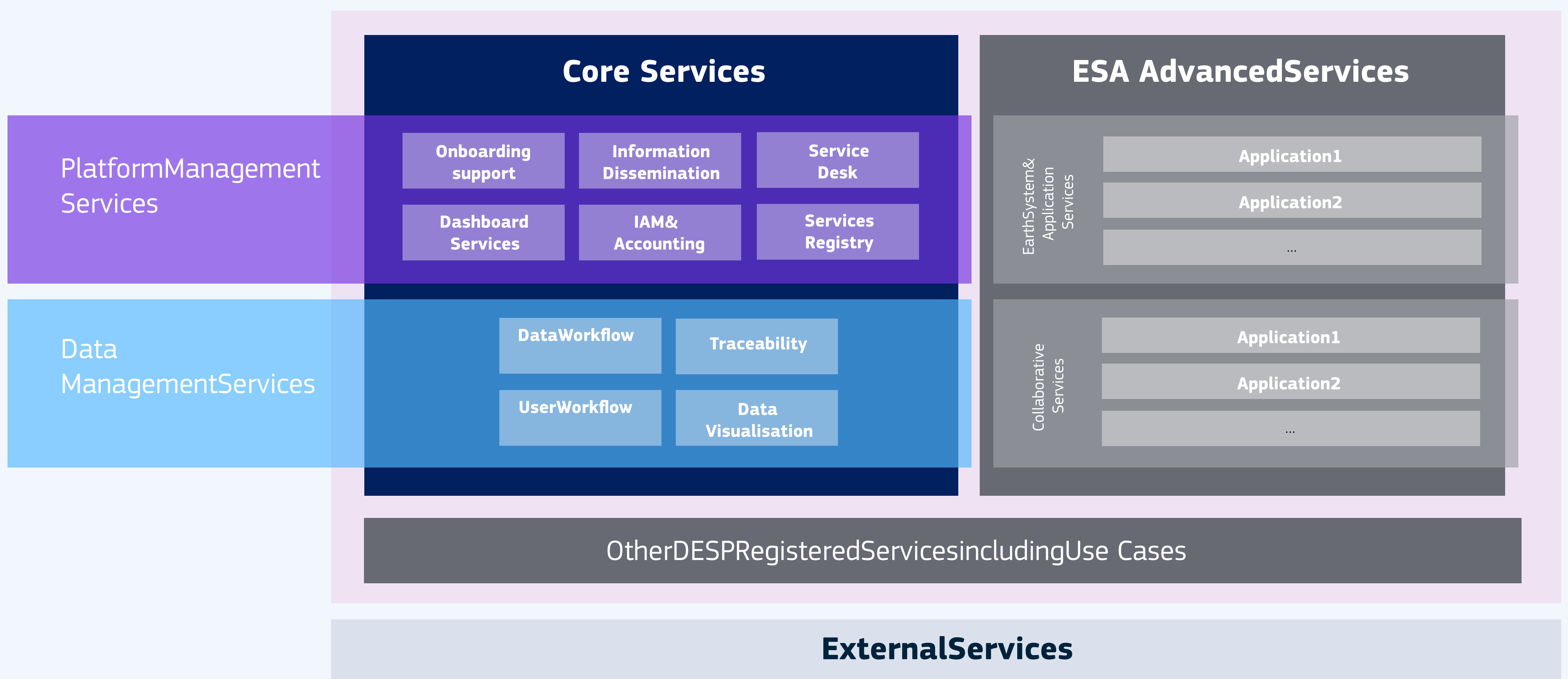
The DestinE Platform is the open and federated digital platform acting as the users single access point for the DestinE ecosystem of tools, data and services.

What can users do?

Through the DestinE Platform, users will access all the free services and features of DestinE, including data workflows and data visualisation services, analytical tools and processing services, as well as on-demand simulations and Earth Observation applications. Users will contribute to co-design the platform by integrating their own data and developing their own applications.

What are the benefits?

The primary purpose of the Platform will be to support evidence-based policy and decision-making.



Source: <https://destination-earth.eu/wp-content/uploads/2023/07/DestinE-Webinar1-DEDL.pdf>

The DestinE Platform is a major component of DestinE, as it will provide a centralised location for accessing and managing the various services and resources that make up the DestinE ecosystem. It includes a wide range of analytical tools and processing services designed to support scientific research including for example advanced data analysis and visualisation capabilities.

These tools allow users to perform complex data processing tasks, such as image analysis, machine learning, and statistical modelling, using the latest algorithms and techniques. This makes it possible to extract valuable insights from large and complex data sets, to develop new models and tools for understanding and managing the Earth system.

Data Lake

What is it?

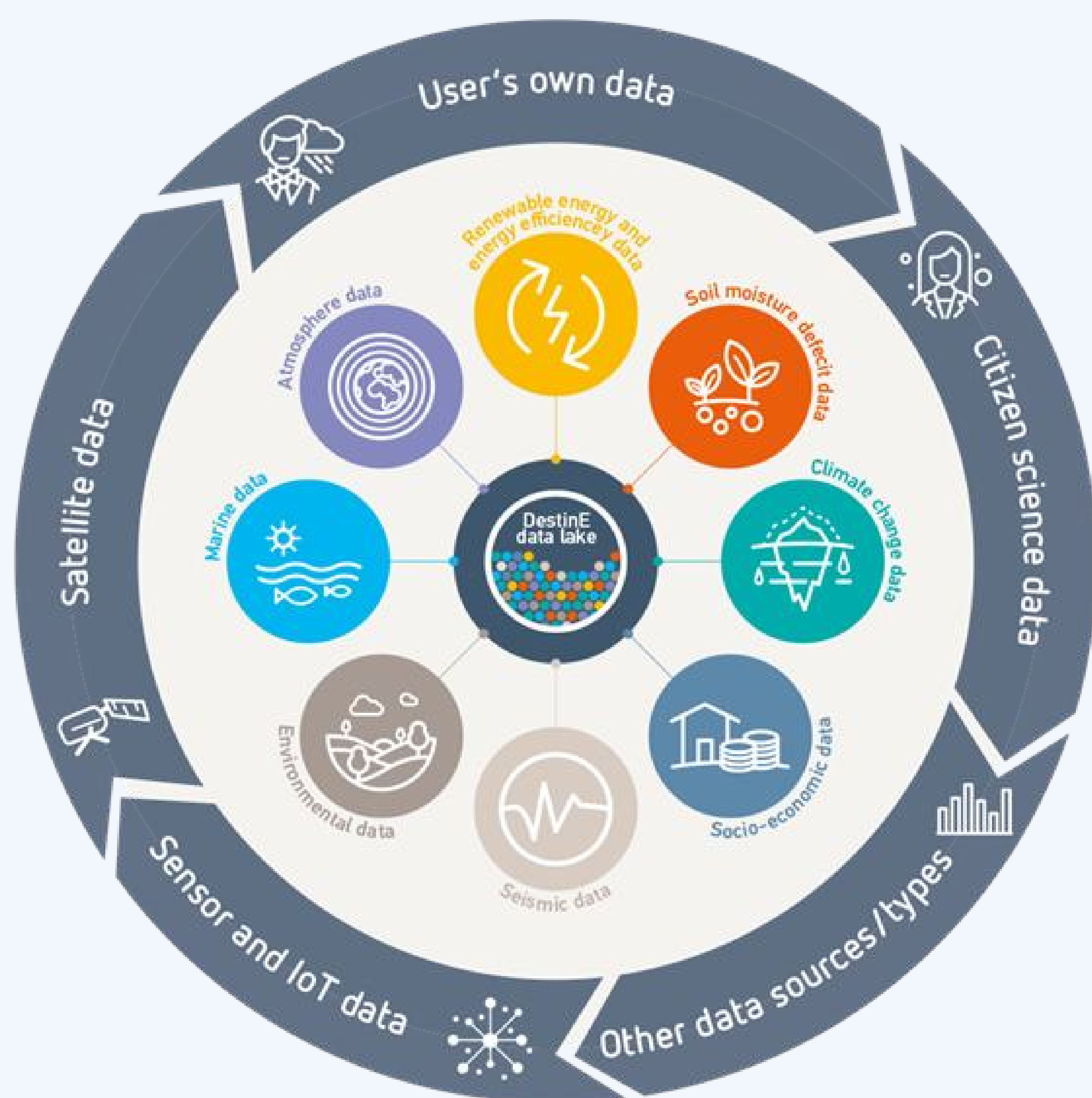
The Data Lake is the DestinE centralised data repository built from geographically distributed data lakes such as Copernicus DIAS, ESA, EUMETSAT, ECMWF as well as other sources like federated data spaces.

What can users do?

Through the DESP, users can seamlessly access all the DestinE datasets included in the Data Lake, regardless of data type and location. In addition to data discovery and access, the Data Lake supports big data processing services, such as near-data processing.

What are the benefits?

The Data Lake fulfils the Big Data processing, data discovery and data access requirements of any DestinE user.



The Data Lake is the DestinE data repository enabling data management capabilities to the DESP. The Data Lake will provide users with on-demand access to a wide range of data sources, including satellite imagery, climate models, and other environmental data.

This data can be used to develop and test new scientific hypotheses, as well as to inform policy decisions related to climate change, natural resource management, and other critical issues.

The Data Lake will constitute a uniquely cohesive, user-focused ecosystem drawing on information from a multitude of different data spaces. This ecosystem will grow alongside the DestinE community, as the more users join and provide their own data, the more the Lake's capabilities and knowledge will increase.

Source: www.eumetsat.int/destination-earth-workshop



Digital twins & Digital Twin Engine

What is it?

The Digital Twin Engine (DTE) is the data handling and software infrastructure needed to operate the DestinE Digital Twins (DT) efficiently in some of the largest supercomputers in the World. The DestinE DTs are digital replicas of the various elements of the Earth system that will eventually contribute to achieving a full digital twin of the Earth.

What can users do?

The DestinE digital twins will allow users to test scenarios to make better decisions about how to mitigate the effects of climate change, assess the socio-economic risk of extreme-weather events.

What are the benefits?

Digital twins significantly enhance Europe's capabilities in understanding and managing extreme environmental events and climate change impacts.

The DTE gives the DestinE DTs a flexible and scalable software architecture, ensuring they work within the DestinE framework and across heterogeneous European supercomputing and data handling infrastructures.

DestinE aims to develop digital twins for multiple Earth System aspects, including Climate Change Adaptation and Weather-Induced Extremes.

Weather-Induced Extremes DT

Developed for ECMWF by a consortium led by Météo France, this DT pushes the boundaries of extreme weather prediction in quality, spatial scale and interactivity. The system equips decision-makers with advanced tools to anticipate the occurrences and impacts of floods, droughts, heat waves and other extreme weather phenomena. This DT will offer on-demand, high-quality global simulation forecasts up to a few days ahead, enabling rapid assessment and prediction of environmental extremes.

Climate Change Adaptation DT

Developed by CSC IT Center for Science on behalf of ECMWF, this DT aims to provide multi-decadal climate simulations with unparalleled level of detail, including specific information for the sectors most affected by climate change. The system will equip decision-makers with advanced, predictive tools to assist in the development and implementation of climate change adaptation strategies.



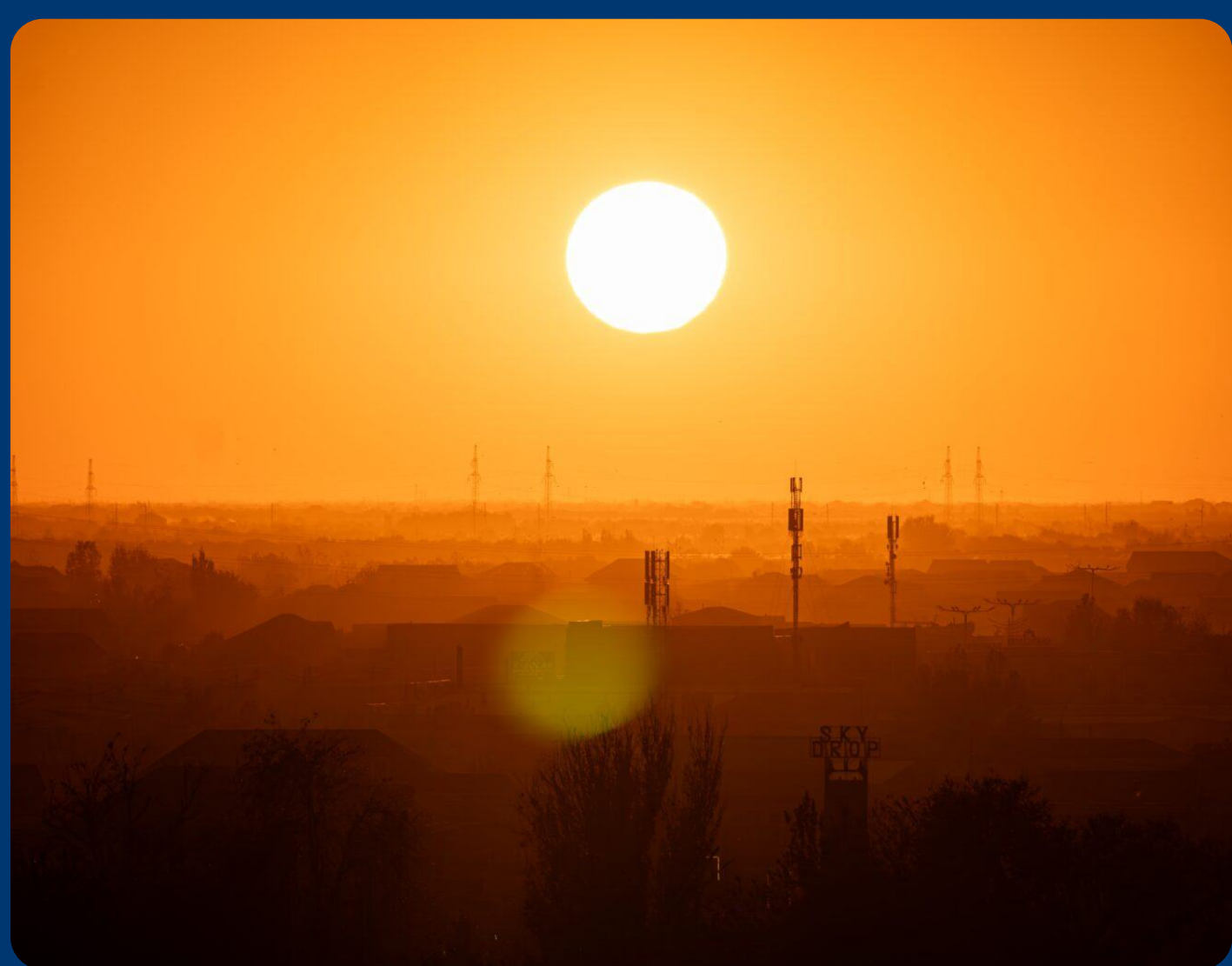
DestinE Use Cases

Demonstrating DestinE's alignment with its users' needs and goals through real-world applications.

The co-design strategy is a cornerstone of the ambitious DestinE initiative. The Use Cases are central to this endeavour, playing an instrumental role in aligning the platform with the distinct objectives and requisites of the user community.

These Use Cases, contracted since the early stages of DestinE illustrate how the Climate Change Adaptation and the Weather-Induced Extremes digital twin are designed by and for the user communities to ensure that DestinE caters real-world applications.

Some of the criteria for selection were credibility and feasibility, alongside a pragmatic roadmap for continued result use.

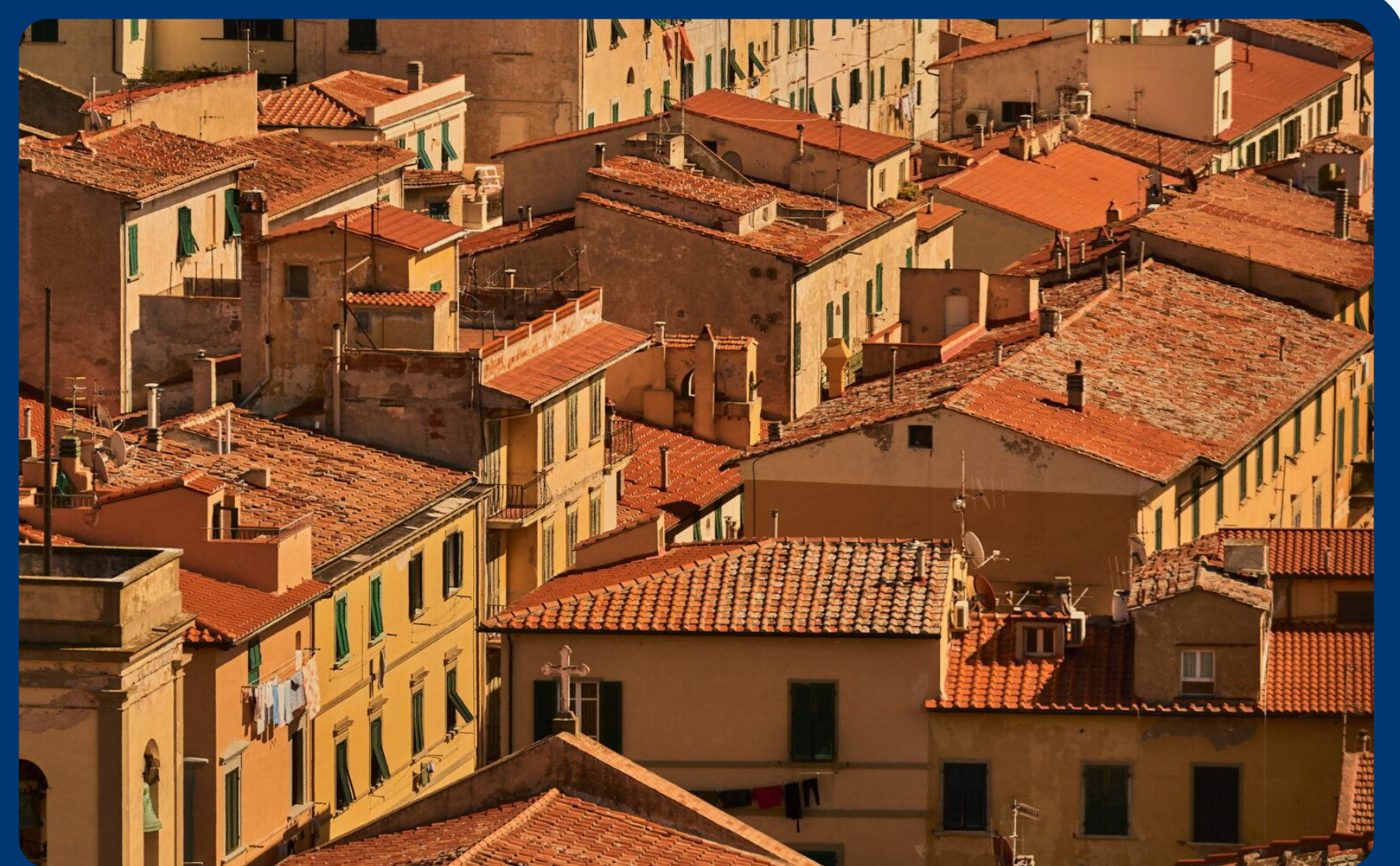


Adaptation Modelling Framework for Heat Stress Exposure

Developing a modelling framework that will facilitate the formulation of adaptation measures addressing heat waves.

Addressing Urban Heat Island Effect

Creating precise heat maps for European cities to address climate change-induced heat stress and enhance urban planning resilience.



Air Quality Forecasting Analysis System

Forecasting air quality in detail, providing insights into pollution exposure, causes and effects and enabling interventions during extreme conditions.

Changes in the Water Cycle

Refining the current hydrology information system to mitigate the effects of water scarcity and water-related disasters.



CITYNEXUS: A novel urban DESP application

Visualising the impacts of urban development initiatives to support data-driven city planning.

DESIDE

Supporting strategic decision-making and policy-makers in decreasing pollution and optimising routes in polar regions.



Destination Renewable Energy (DRE)

Simulating the physical systems of solar and wind production to support decision-making on energy use, trade, and storage.

Disaster Risk Mitigation & Climate Adaptation

Enhancing delta regions' preparedness for compound flooding events with mitigation strategies, increasing safety and reducing flood impacts.





Global Fish Tracking System (GFTS)

Predicting the behaviour and changes in fish population to support data-driven conservation policies.

Improved Energy System Modelling

Bolstering electric systems in facing climate change-induced extreme weather events and renewable energy transition challenges.



Italy Drought 2022

Supporting a more accurate assessment of the impact of ongoing droughts and identification of vulnerable areas.

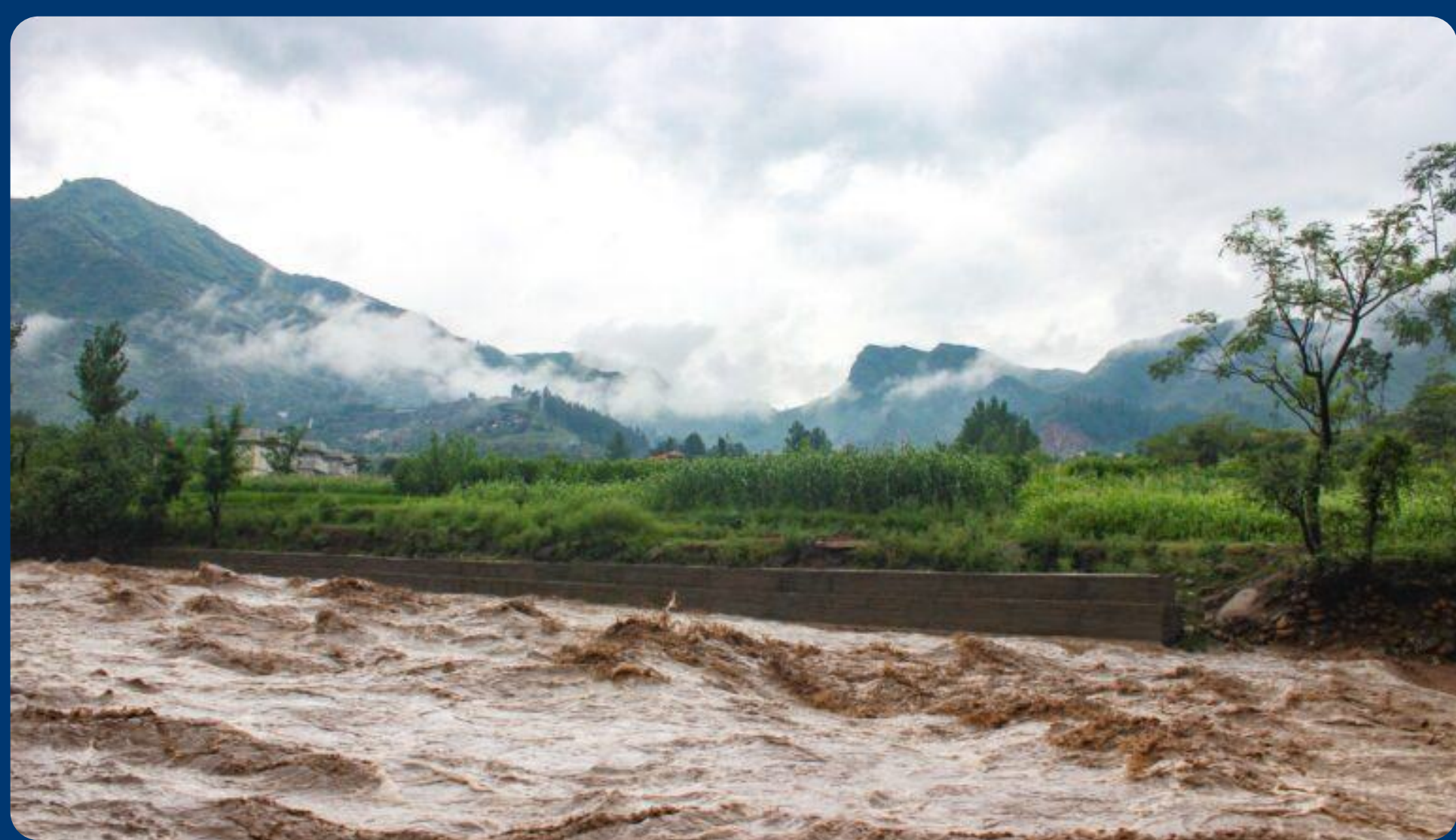
More Resilient City Planning

Producing a heat stress index to support better decision-making and data-driven adaptation policies in cities.



Onshore & Offshore Wind Energy Information

Exploiting wind power efficiently to guarantee reliability and security of the electricity system.

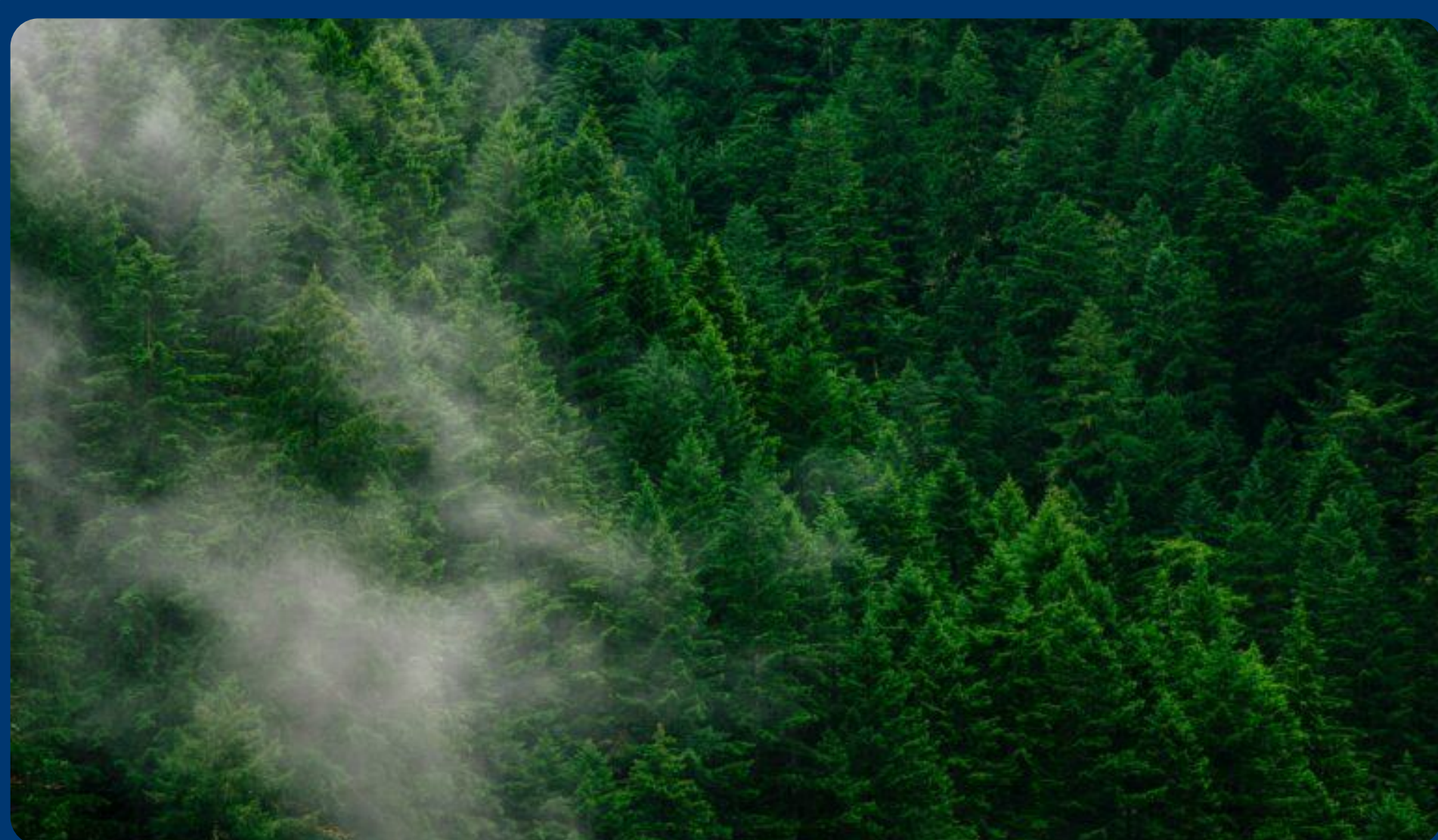


Pakistan Flood 2022

Enabling enhanced flood risk management for more accurate and targeted mitigation measures.

Simulating the Future of Extreme Events

Providing accurate statistics of hydrological extremes to estimate the impacts of climate change.

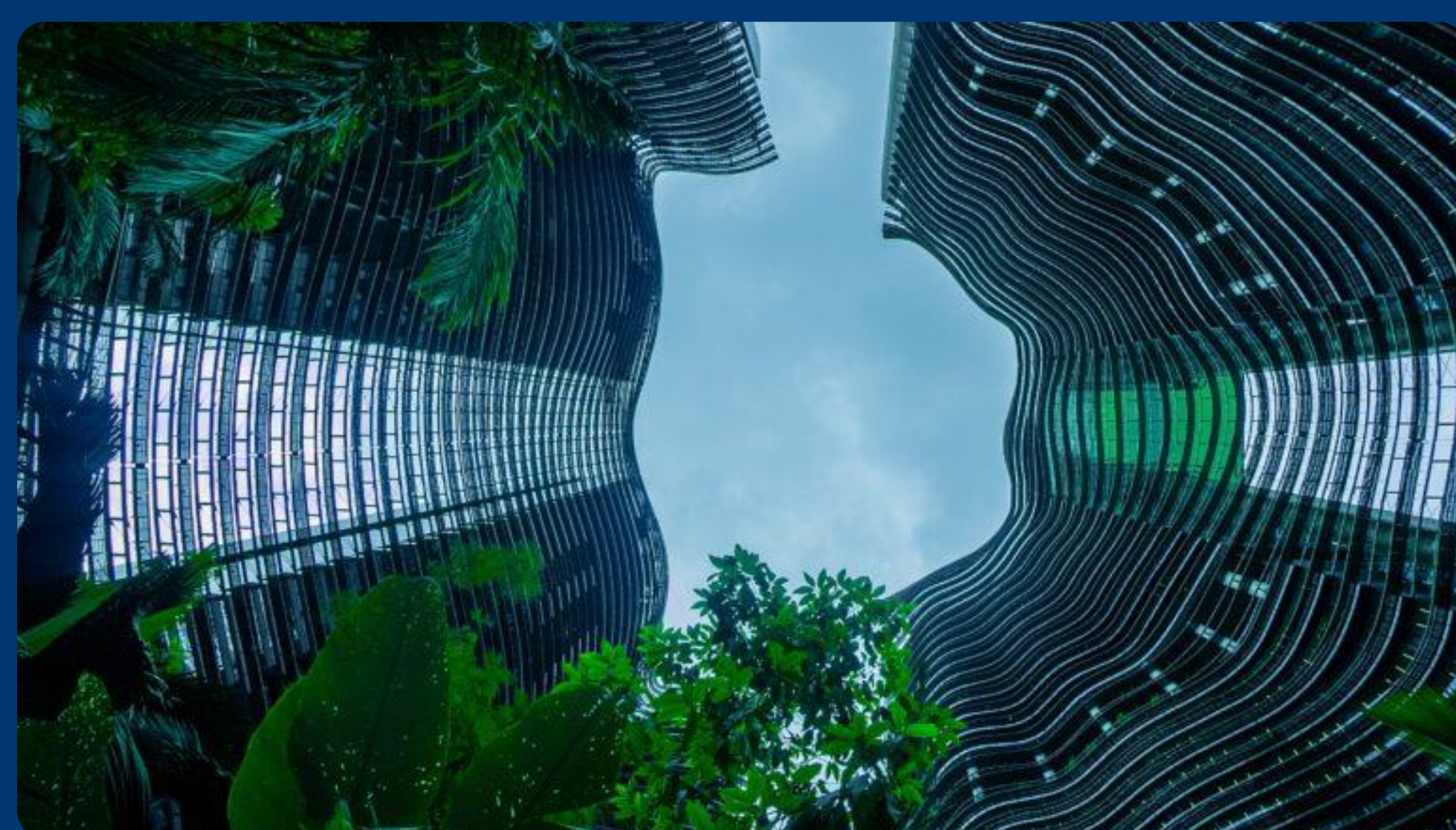


Sustainable Forest Management

Providing harvestability maps and climate change data to support sustainable forest management, benefiting users and enhancing coordination within the sector.

UrbanSquare

Assessing climate-related challenges faced by cities such as heat waves, flooding, and air pollution to support urban climate resilience.



Wildfires Evolution

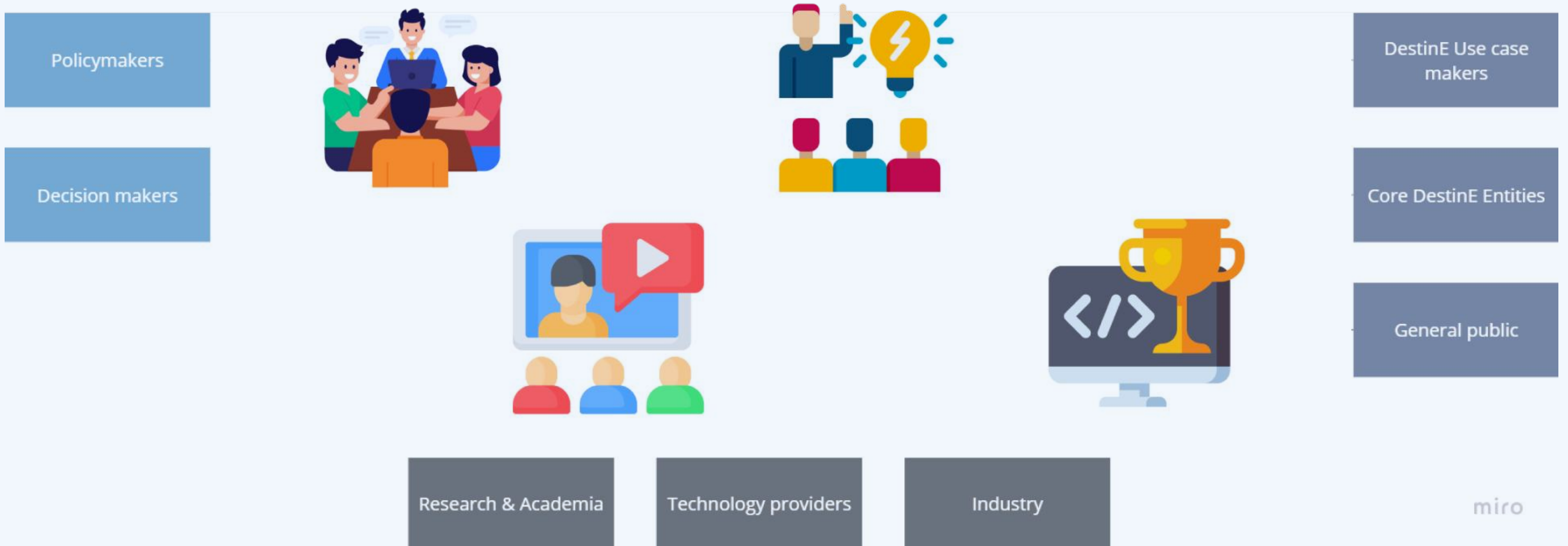
Wildfires Evolution: Providing tools to assess present and future wildfire risks and potential solutions in a changing climate.

DestinE

Use Cases



DestinE Community



co-design, validate, develop and use the DestinE Components' capabilities in an open, inclusive and transparent environment

Source: <https://destination-earth.eu/wp-content/uploads/2023/07/Community.pdf>

Building a DestinE community is crucial to identify user needs and preferences, tap into the knowledge and expertise of individuals from different fields or industries, and develop applications that meet user needs.

The DestinE Community welcomes individuals, from policy makers to DestinE use cases, core entities, the general public, research and academia, technology providers, decision makers and industry representatives from various fields to guide the development and sustainability of the system.

The community will be a network of users that interact continuously with developers, stakeholders, partners, and other users. Its role is to enhance the development and improvement of the DestinE initiative capabilities while encouraging collaboration. By tapping into existing communities, the DestinE community aims to create a wide community of practice of Earth System Digital Twins.

Members of the community can follow the discussion through the DestinE website and stay informed of any co-development opportunities that may arise in the future and can participate in events and discuss various topics related to the digital twins, the Digital Twin Engine, the Lake, and the Platform. Contribute to the impact of DestinE. Help us turn this ambitious initiative into a reality.

Join the DestinE community!

Become a User

DestinE data is designed to empower users by equipping them with the enhanced tools to generate actionable information, share knowledge, engage in research, validate models, and test hypotheses concerning the Earth system. This includes a comprehensive exploration of the intricate and interconnected roles that both the environment and humanity will play in shaping the future of our planet.

There are several ways to become a DestinE User including;

1. Join Our Online Community: Keep an eye out for our forthcoming online community launch on our website, providing a platform for collaborative engagement. Here, you can explore:
 - a. Events like Collaborative Design Sessions, Hackathons, Webinars and Conference Sessions.
 - b. Project News
 - c. DestinE Journal Articles
2. Subscribe to our newsletter to receive notifications of future opportunities directly to your inbox.
3. Participate in Our Upcoming Events: Gain firsthand insights into the latest developments and updates regarding DestinE from the entities and contractors involved in funded projects.

“A community of users that we aim to create in order to guide the development and sustainability... where the continuous interactions amongst users, developers, stakeholders, partners will enhance the development of and improvement of DestinE’s capabilities but also catalyse cross sectoral collaborations” - Eleni Karachaliou, Community Manager



Become a Technology Provider

Becoming a technology provider for Destination Earth is a great way to contribute to pushing the boundaries in our understanding of climate change and extreme weather. By supporting the development and implementation of DestinE with tools and resources (such as data processing, algorithms and storage solutions), technology providers will reach new markets, access a wealth of tools and data and be able to test and develop their products and services. Through the following steps, you can get involved and help make DestinE a reality.

1. Visit the Destination Earth Joint website and subscribe to the newsletter. This will keep you up-to-date on the latest developments and co-development opportunities.
2. Apply to participate in our events and workshops. This can be a great way to network with other technology providers and to learn more about how you can contribute.
3. Submit proposals for funding. As a technology provider, you may be eligible for funding to support your contributions to the initiative. Keep an eye out for calls for proposals related to Destination Earth and be sure to submit your ideas.



Procurement Opportunities

The entities implementing DestinE offer numerous opportunities for co-development. The private sector and other interested organisations are key collaborators and enablers, playing a crucial role in developing components, implementing use cases and enriching the system through services and applications.

Within this inclusive approach, diverse stakeholders are invited to contribute to the creation of DestinE.

Organisations interested in taking an active role in the implementation of DestinE can monitor and respond to procurement opportunities as soon as they are made public by ESA, ECMWF or EUMETSAT. These procurement opportunities will be showcased on the DestinE website:

www.destination-earth.eu/procurement.

Interested organisations can also monitor the agencies' respective procurement websites for opportunities related to DestinE. ECMWF offers a dedicated DestinE procurement page, and ESA and EUMETSAT also have established a dedicated portal for publishing relevant procurement notices.

Join the Community

Putting cutting-edge science and technology in the hands of scientists and policymakers to help them make the right decisions.

Putting cutting-edge science and technology in the hands of scientists and policymakers to help them make the right decisions.

The DestinE Community places the active engagement of users at its core, nurturing collaboration in the co-design and development of DestinE. This collaborative approach amplifies DestinE's capabilities, equipping it to address complex environmental challenges with innovation and precision.



Take the opportunity now to be a part of a dynamic community that shares knowledge, co-creates solutions, and unlocks innovative pathways for addressing shared challenges. Become part of the DestinE Community!

